
ALGEBRA-SUMMER

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0.1 Intro

Theorem 0.1.1 (Division Algoritm). Suppose $a, b \in \mathbb{Z}$ s.t. $b > 0$. Then there are unique $q, r \in \mathbb{Z}$ s.t. $a = bq + r$ and $0 \leq r < b$

Proof. Let $a, b \in \mathbb{Z}$ s.t. $b > 0$. Consider $A = \{a - bx \mid x \in \mathbb{Z}\} \cap \mathbb{N}$

By the Well Ordering Principle since A has all non-negative terms it has a minimum lets call this r and call the associated $x \in \mathbb{Z}$, q .

Thus, $r = a - bq \implies a = bq + r$.

For contradiction, assume $r \geq b$.

$\implies r = b + m$ s.t. $m \geq 0$

$\implies b + m = a - bq \implies m = a - b(q + 1) \in A \implies m < r = \text{minimum} \implies \Leftarrow$ Thus $0 \leq r < b$

Fix $q_1, q_2, r_1, r_2 \in \mathbb{Z}$ s.t. $a = bq_1 + r_1 = bq_2 + r_2$ and $0 \leq r_1, r_2 < b$. WLOG let $r_1 \geq r_2$

$\implies bq_1 + r_1 = bq_2 + r_2 \implies r_1 - r_2 = b(q_2 - q_1) \implies r_1 - r_2$ is a multiple of b

Since $0 \leq r_1, r_2 < b \implies 0 \leq r_1 - r_2 < b$

Since $r_1 - r_2$ is strictly less than b the only multiple of b it could be is 0 thus $r_1 = r_2$

$\implies q_1 = q_2$ □

Lemma 0.1.2. (Bezout's Lemma) Suppose $a, b \in \mathbb{Z} \setminus \{0\}$. Then there are $x, y \in \mathbb{Z}$ s.t. $ax + by = \gcd(a, b)$.

Proof. Let $a, b \in \mathbb{Z} \setminus \{0\}$. Consider $A = \{ax + by \mid x, y \in \mathbb{Z}\} \cap \mathbb{N}$

By the Well Ordering Principle since A has all non-negative terms it has a minimum lets call this d and call the associated $x, y \in \mathbb{Z}$, u and v .

By Theorem 0.1.1 $\exists q, r \in \mathbb{Z}$ s.t. $a = dq + r$ and $0 \leq r < d$

Case 1: $r = 0 \implies d \mid a$

Case 2: $0 < r < d \implies r = a - dq \implies r = a - (au + bv)q = a(1 - uq) + b(-vq) \in A$

$\implies r < d = \text{minimum} \implies \Leftarrow$

Thus $d \mid a$ and similarly $d \mid b$

Let $c \in \mathbb{N}$ s.t. $c \mid a$ and $c \mid b$

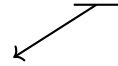
$\implies a = cm$ and $b = cn$ for $m, n \in \mathbb{Z}$

$d = au + bv = c(mu) + c(mv) = c(mu + mv) \implies c \mid d \implies d = \gcd(a, b)$ □

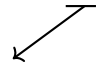
Algorithm 0.1.3. (Euclidean Algorithm)

Find $\gcd(272, 571)$ and $x, y \in \mathbb{Z}$ s.t. $272x + 571y = \gcd(272, 571)$

$$571 = 272 \cdot 2 + 27 \quad (1)$$



$$272 = 27 \cdot 10 + 2 \quad (2)$$



$$27 = 2 \cdot 13 + 1 \quad (3)$$

Last term before remainder is zero is gcd

$$\gcd(272, 571) = 1$$

$$2 = 272 - 27 \cdot 10$$

Use equation (2)

$$= 272(21) - 571(10)$$

$$1 = (571 - 272 \cdot 2) - 13(272(21) - 571)$$

Use equation (3) with (2)

$$= 571(131) + 272(-275)$$

1 Groups

1.1 Intro

Definition 1.1.1. A group is a set G together with a binary operation $*$: $G \times G \rightarrow G$ s.t.

- **(Closure)** For all $a, b \in G$ we have $a * b \in G$
- **(Identity)** There is $e \in G$ s.t. for all $a \in G$ $a * e = e * a = a$
- **(Inverses)** For all $a \in G$ there is an $a^{-1} \in G$ s.t. $a * a^{-1} = a^{-1} * a = e$
- **(Associativity)** For all $a, b, c \in G$ we have $a * (b * c) = (a * b) * c$

Definition 1.1.2. Given a group G and $g \in G$ we say the order of g is $n \in \mathbb{N}$ if n is the minimum value s.t. $g^n = g * g * \dots * g = e$
Write as: $\text{ord}(g) = |g| = n$

1.1.1 Exercises

Example 1.1.3. Let $G = \mathbb{R} \setminus \{-1\}$ and define $a * b = a + b + ab$. Show $(G, *)$ is an Abelian group

- (Abelian) For all $a, b \in G$ $a * b = a + b + ab = b + a + ba = b * a$
- (Closure) $\checkmark$
- (Identity) For all $a \in G$ $a * 0 = 0 * a = 0$
- (Inverses) For all $a \in G$ let $b = \frac{-a}{a+1}$.

$$\text{Thus, } a * b = b * a = a + b + ab = a + b(a+1) = \frac{-a}{a+1}(a+1) = a - a = 0$$

- (Associativity) For all $a, b, c \in G$

$$a * (b * c) = a * (b + c + bc) = a + b + c + bc + ab + ac + abc = (a + b + ab) * c = (a * b) * c$$

Therefore, G is an Abelian group

Example 1.1.4. Find all $x \in \mathbb{Z}$ s.t. $5x \equiv 4 \pmod{9}$

$$5x \equiv 4 \pmod{9}$$

$$x \equiv 8 \pmod{9}$$

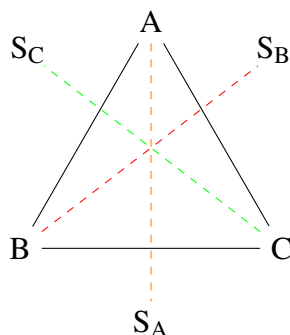
$$x = 9n + 8$$

$$2 * 5 = 10 \equiv 1 \pmod{9}$$

$$n \in \mathbb{Z}$$

1.2 Dihedral Groups

1.2.1 Rigid Motions of an Equilateral Triangle



Rotations

$r = 120^\circ$ rotation

$r^2 = 240^\circ$ rotation

$r^3 = 360^\circ$ rotation $= e = 0^\circ$ rotation

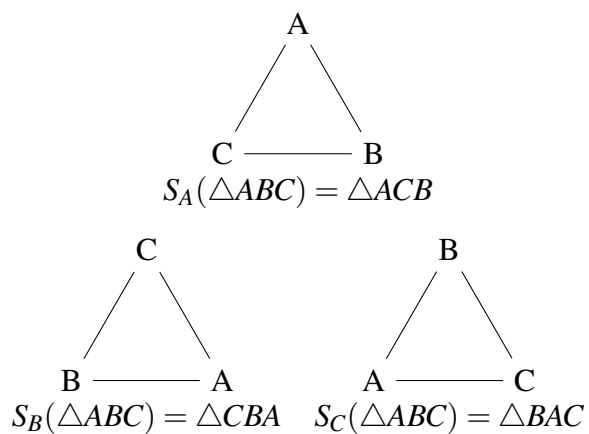
Reflections

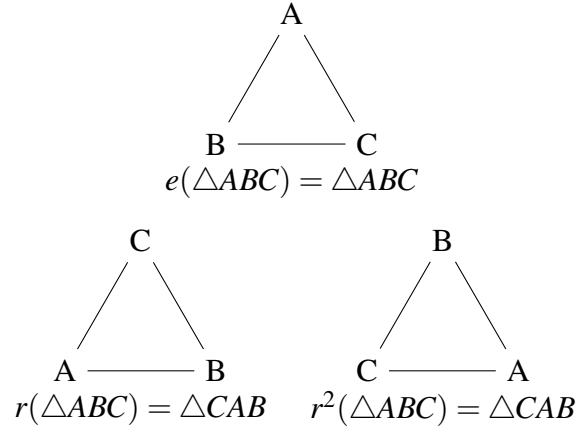
$S_A : \triangle ABC \rightarrow \triangle ACB$

$S_B : \triangle ABC \rightarrow \triangle CBA$

$S_C : \triangle ABC \rightarrow \triangle BAC$

Transformation Diagrams





Operations

$$\begin{aligned} r \circ S_A(\triangle ABC) &= r(\triangle ACB) = \triangle BAC = S_C(\triangle ABC) & r \circ S_A &= S_C \\ r^2 \circ S_A(\triangle ABC) &= r^2(\triangle ACB) = \triangle CBA = S_B(\triangle ABC) & r^2 \circ S_A &= S_B \end{aligned}$$

Now we only need S_A so lets call this S

Thus we only need S, r, r^2, e

Also $S^2 = r^3 = e$

How S, r, r^2 Commute

$$\begin{aligned} r \circ S(\triangle ABC) &= r(\triangle ACB) = \triangle BAC = S \circ r^2 \\ r^2 \circ S(\triangle ABC) &= r^2(\triangle ACB) = \triangle CBA = S \circ r \end{aligned}$$

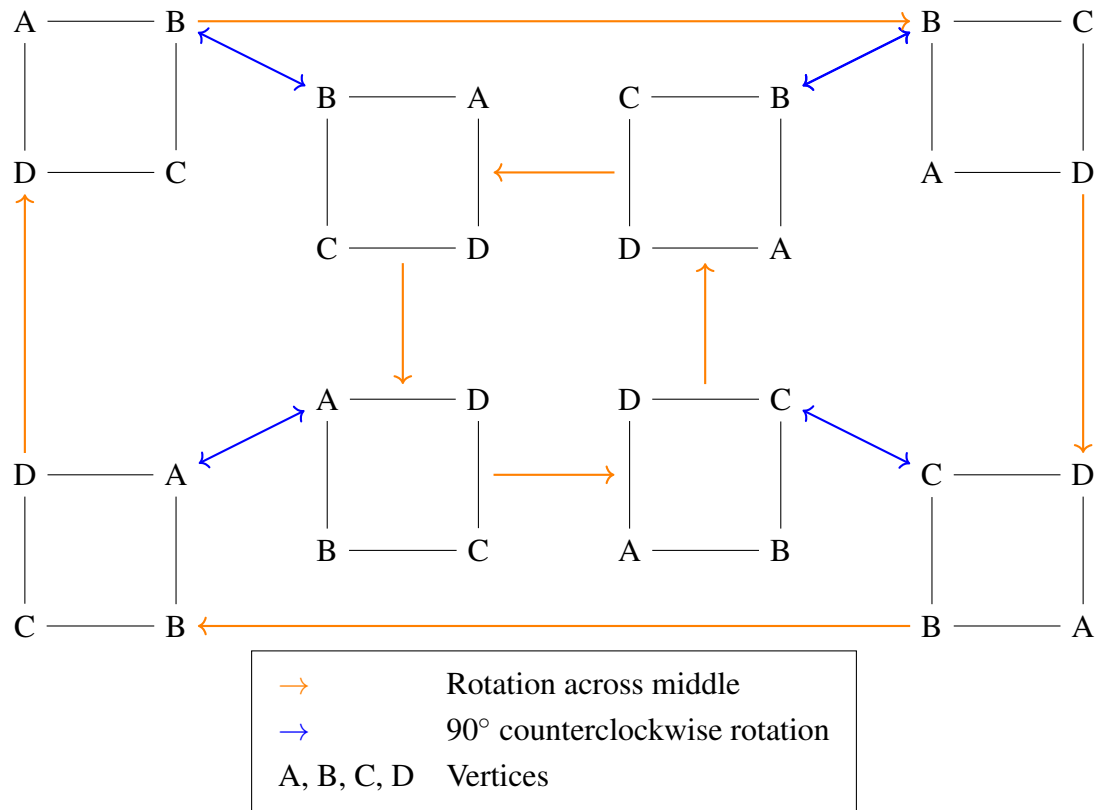
Definition 1.2.1. The group of symmetries of an equilateral triangle is $D_3 = \{e, r, r^2, S, Sr, Sr^2\}$, where $r^3 = S^2 = e$, $rS = Sr^2$

Dihedral group of order $2 \cdot 3 = 6$

Compact Notation $D_3 = \langle r, s^1 \mid r^3 = S^2 = e, rS = Sr^2 \rangle$

¹The generators of D_3 meaning you can create all other operations from them

1.2.2 Symmetry Group for a Square



Rotations

$r = 90^\circ$ rotation
 $r^2 = 180^\circ$ rotation
 $r^3 = 270^\circ$ rotation
 $r^4 = 360^\circ$ rotation = $e = 0^\circ$ rotation

Reflections

$$s : \square ABCD \rightarrow \square BCDA$$

How does s, r Commute

From the diagram we see that $sr = r^3s \implies rs = sr^3$

Definition 1.2.2. The group of symmetries of a square is $D_4 = \{e, r, r^2, r^3, s, sr, sr^2, sr^3\}$

Dihedral group of order $2 \cdot 4 = 8$

$$D_4 = \langle s, r \mid r^4 = s^2 = e, sr = r^3s \rangle$$

1.2.3 Symmetry Group of regular n-gon

Rotation: $r = \frac{360^\circ}{n} \circlearrowleft (e, r, r^2, \dots, r^{n-1})$

Reflection: $s = \text{reflection across middle } (s, sr, sr^2, \dots, sr^{n-1})$

Claim: $rs = sr^{n-1}$.

Proof. We only need 3 points since our shape is defined by the order of those 3 points

$$r \circ s(n, 1, 2) = r(2, 1, n) = (1, n, n-1)$$

$$s \circ r^{n-1}(n, 1, 2) = s(n-1, n, 1) = (1, n, n-1)$$

$$\text{Thus } rs = sr^{n-1}$$

□

Definition 1.2.3. The dihedral group D_n is the symmetry group of a regular n-gon.

$$D_n = \langle s, r \mid r^n = s^2 = e, rs = sr^{n-1} \rangle$$

Claim: For all $1 \leq m \leq n-1$ we have $r^m s = sr^{n-m}$.

Proof. Base Case: $m = 1 \implies rs = sr^{n-1} \checkmark$

Inductive Step: Assume for $k \geq 1$ the statement is true

Consider:

$$r^{k+1}s = rr^k s$$

$$= rsr^{n-k}$$

$$= sr^{n-1}r^{n-k}$$

$$= sr^n r^{n-(k+1)}$$

$$= sr^{n-(k+1)}$$

Apply inductive hypothesis on $r^k s$

□

1.2.4 Exercises

Example 1.2.4. Simplfy

- $(sr^5)r(sr^3)$ in D_4

$$= (sr)r(rs)$$

$$= e$$

- $(sr^4)(sr^2)$ in D_6

$$= (r^2s)(sr^2) \quad \text{Use } r^m s = sr^{n-m}$$

$$= r^4$$

- $r^3(sr)$ in D_4

$$= r^3(r^3s)$$

$$= r^2s$$

Example 1.2.5. The Cayley table of the D_3 can be written

	e	r	r^2	s	rs	r^2s
e	e	r	r^2	s	rs	r^2s
r	r	r^2	e	rs	r^2s	s
r^2	r^2	e	r	r^2s	s	rs
s	s	r^2s	rs	e	r^2	r
rs	rs	s	r^2s	r	e	r^2
r^2s	r^2s	rs	s	r^2	r	e

1.3 Symmetric Groups

Definition 1.3.1. Given a set X , a permutation of X is a bijection $X \rightarrow X$
 $S_X = \{ \sigma : X \rightarrow X \mid \sigma \text{ is a bijection} \}$

Lemma 1.3.2. For all sets $X \neq \emptyset$, S_X is a group

Proof.

- Closure: Suppose $\sigma, \mu \in S_X$
 Consider $\sigma \circ \mu$ since it is a composition of bijections it is also a bijection.
 Also functions go from $X \rightarrow X$ thus $\sigma \circ \mu$ is the same.
 Therefore, the composition is a bijection that is mapped from $X \rightarrow X$ thus $\sigma \circ \mu \in S_X$
- Identity: Consider $e : X \rightarrow X$ s.t. $e(x) \mapsto x$
 Thus $e \in S_X$ and for all $\sigma \in S_X$ $\sigma \circ e(x) = \sigma(x)$
 Thus $\sigma \circ e = \sigma$. Also, it follows that $e \circ \sigma = \sigma$
- Inverses: Since S_X has only bijective functions all bijective functions have inverses.
- Associativity: We have associativity by the fact that function composition has associativity.

□

Definition 1.3.3. The permutation group on n -objects S_n is the S_X with $X = [n]$
 Observe $|S_n| = n!$

Example 1.3.4. $\sigma, \mu \in S_4$

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \mu = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix}$$

$$\sigma^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \mu^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix}$$

$$\sigma\mu = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \mu\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix}$$

1.3.1 Cycle Notation

Example 1.3.5. $\sigma, \mu \in S_5$

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 4 & 2 & 5 & 1 \end{pmatrix} \mu = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 5 & 2 & 1 & 3 \end{pmatrix}$$

$$\sigma = (13245) \mu = (14)(253)$$

Example Calculations:

$$\sigma\mu = (13245)(14)(253)$$

$$= (152)(34)$$

$$\mu\sigma = (14)(253)(13245)$$

$$= (12)(354)$$

$$\mu^2 = (14)(253)(14)(253)$$

$$= (14)^2(253)^2 = (235)$$

Lemma 1.3.6. *Disjoint cycles are commutative*

Proof. Let α, β be disjoint cycles of $[n]$

Let $\alpha = (a_1, a_2, \dots, a_k)$, and $\beta = (b_1, b_2, \dots, b_j)$

Let $x \in [n]$

Consider: $\alpha\beta(x), \beta\alpha(x)$

If $x \in \alpha$ then $x \notin \beta \implies \beta(x) = x \implies \alpha\beta(x) = \alpha(x) = a_i$
 $\implies \beta\alpha(x) = \beta(a_i) = a_i$ since $a_i \notin \beta$

Use similar reasoning for if $x \in \beta$

$\implies \alpha\beta = \beta\alpha$

□

Lemma 1.3.7. *Every $\sigma \in S_n$ can be written as disjoint cycles*

Lemma 1.3.8. *The order of an m -cycle is m*

Lemma 1.3.9. Suppose G is a group and $\text{ord}(g) = n$. Then $g^m = e \iff n \mid m$

Proof. Let G be a group and $\text{ord}(g) = n$

($\Leftarrow$)

Assume $n \mid m$

$$\implies m = nk, k \in \mathbb{Z}$$

$$g^m = g^n k = (g^n)^k = (e)^k = e$$

($\Rightarrow$)

Assume $g^m = e$ and use the division algorithm (0.1.1) to write $m = nq + r$ s.t. $0 \leq r < n$

$$e = g^m = g^{nq+r} = (g^n)^q g^r = g^r \implies r = 0 \implies n \mid m$$

□

Theorem 1.3.10. If $\sigma_1, \sigma_2, \dots, \sigma_k$ are disjoint cycles of length $a_1, a_2, \dots, a_k$ and $\sigma = \sigma_1 \sigma_2 \dots \sigma_k$. Then $\text{ord}(\sigma) = \text{lcm}(a_1, a_2, \dots, a_k)$.

Proof. Let $m = \text{ord}(\sigma)$, $l = \text{lcm}(a_1, a_2, \dots, a_k)$

Since l is a lcm it is also a multiple so $l = a_j q_j$ for $1 \leq j \leq k$ where $q_j \in \mathbb{Z}$

$$\sigma^l = \sigma_1^l \sigma_2^l \dots \sigma_k^l$$

We can do this because of Lemma 1.3.6

$$= (\sigma_1^{a_1})^{q_1} \dots (\sigma_k^{a_k})^{q_k}$$

$$= (1)$$

Apply Lemma 1.3.8

$$\implies m \mid l$$

Also, $(1) = \sigma^m = \sigma_1^m \dots \sigma_k^m$ since they are disjoint none of them are inverses of each other

$$\implies \sigma_1^m = \dots = \sigma_k^m = (1)$$

$$(\text{Lemma 2.1.9}) \implies a_1 \mid m \dots a_k \mid m \implies l \mid m$$

Thus, since $l \mid m$ and $m \mid l$ $m = l$

□

Example 1.3.11.

- Find an element of order 12 in S_7
 $(1234)(567) \in S_7$ by Theorem 1.3.10 $\implies$ the order is $3 \cdot 4 = 12 = \text{lcm}(3,4)$
- Possible orders of elements in S_5
 - 1: (1) , $5=1+1+1+1+1$
 - 2: (12) , $5=2+1+1+1+1$ or $(12)(34)$, $5=2+2+1$
 - 3: (123) , $5=3+1+1$
 - 4: (1234) , $5=4+1$
 - 5: (12345) , $5=5$
 - 6: $(12)(345)$, $5=2+3$

1.3.2 Exercises

Example 1.3.12. Write the following as the product of disjoint cycles and find the order.

- $(1358)(378)(125) \in S_8$
 (128537) order=6
- $(1264)^{-1}(364)^2 \in S_6$
 $= (1462)(346)$
 $= (142)(36)$ order=6

Example 1.3.13. Find all Possible orders of elements in S_7

- 1: (1) , $7=1+1+1+1+1+1+1$
- 2: (12) , $7=2+1+1+1+1+1$ or $(12)(34)$, $7=2+2+1+1+1$ or $(12)(34)(56)$, $7=2+2+2+1$
- 3: (123) , $7=3+1+1+1+1$ or $(123)(456)$, $7=3+3+1$
- 4: (1234) , $7=4+1+1+1$ or $(1234)(56)$, $7=4+2+1+1$
- 5: (12345) , $7=5+1+1$
- 6: (123456) , $7=6+1$ or $(123)(45)$, $7=3+2+1+1$, or $(123)(45)(67)$, $7=3+2+2+1$
- 7: (1234567) , $7=7$
- 10: $(12345)(67)$, $7=5+2$
- 12: $(1234)(567)$, $7=4+3$

Example 1.3.14. What is the largest possible order of an element in S_{10} ?

cycle type	order
(5,1,1,1,1)	5
(5,2,1,1,1)	10
(5,2,2,1)	10
(5,3,1,1)	15
(5,3,2)	30
(5,4,1)	20
(5,5)	5
(6,1,1,1,1)	6
(6,2,1,1)	6
(6,3,1)	6
(6,4)	12
(7,1,1,1)	7
(7,2,1)	14
(7,3)	21
(8,1,1)	8
(8,2)	8
(9,1)	9
(10)	10

Max=30

1.4 General Results on Groups

Lemma 1.4.1. *Suppose G is a group. Then*

- (1) *The identity is unique*
- (2) *Inverses are unique*
- (3) *For every $a, b, c \in G$ there is a unique $x \in G$ s.t. $ax = b$*
- (4) *If $ab = ac$ then $b = c$, similarly if $xy = zy$ then $x = z$*
- (5) $(ab)^{-1} = b^{-1}a^{-1}$
- (6) *If $(ab)^2 = a^2b^2$ then $ab = ba$*
- (7) *If $\text{ord}(g) = 2$ for all $g \in G$ then G is abelian*

• *Proof.* (3) Let $ax = b \implies x = a^{-1}b$

Suppose $x, y \in G$ s.t. $ax = b$ and $ay = b$

$$ax = ay \implies x = y \quad \square$$

• *Proof.* (6) Assume $(ab)^2 = a^2b^2$

$$ab = (a^{-1}a)(ab)(bb^{-1}) = a^{-1}a^2b^2b^{-1} = a^{-1}(ab)^2b^{-1} = (a^{-1}a)(ba)(bb^{-1}) = ba \quad \square$$

• *Proof.* Assume $\text{ord}(g) = 2$ for all $g \in G$

$$\implies g = g^{-1} \text{ for all } g \in G$$

Let $a, b \in G$. Consider ab

$$ab = (ab)^{-1} = b^{-1}a^{-1} = ba \quad \square$$

Definition 1.4.2. Given a group G and a subset $H \subseteq G$, we say H is a subgroup $H \leq G$ if

- The identity of G is in H
- For all $h \in H, h^{-1} \in H$
- For all $a, b \in H, ab \in H$

Theorem 1.4.3. Suppose that G is a group and $\emptyset \neq H \subseteq G$
 then $H \leq G \iff$ for all $a, b \in H, ab^{-1} \in H$

Proof.

($\implies$) Assume $H \leq G$

Let $a, b \in H$

By Definition 1.4.2 $\implies b^{-1} \in H \implies ab^{-1} \in H$

($\impliedby$) Assume for all $a, b \in H, ab^{-1} \in H$

- (Identity) Consider $a \in H$

$$\implies aa^{-1} = e \in H$$

- (Inverses) $\checkmark$

- (Closure) Consider $a, b^{-1} \in H$

$$\implies ab \in H$$

□

1.4.1 Subgroup Lattice

Subgroup of Order 4

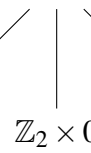
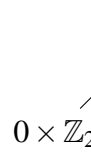
$$\mathbb{Z}_4 = \{0, 1, 2, 3\}$$



$$\begin{array}{c} | \\ 0 = \{0\} \end{array}$$

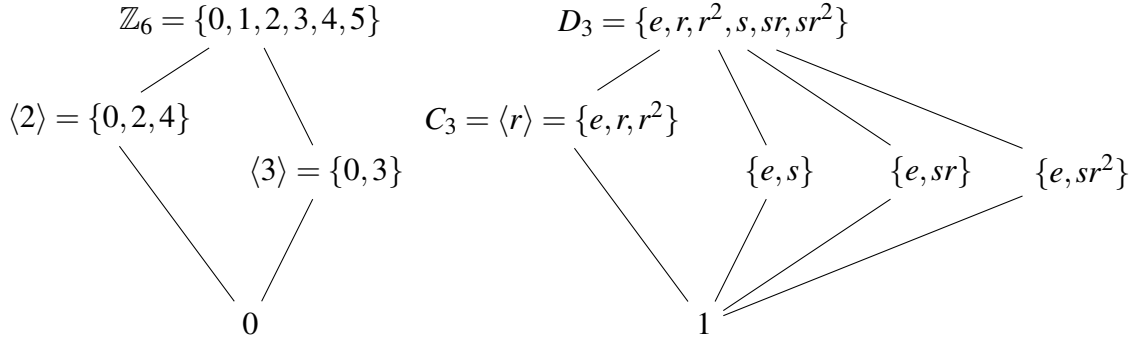
$$0 = \{0\}$$

$$\mathbb{Z}_2 \times \mathbb{Z}_2 = \{(0, 0), (1, 0), (0, 1), (1, 1)\}$$



$$0 = \{(0, 0)\}$$

Subgroup of Order 6



Definition 1.4.4. Given a group G and a subset $S \subseteq G$

- The centralizer of S in G : $C_G(S) = \{g \in G \mid gs = sg, \forall s \in S\}$
- The center of G : $(C_G(G)) = Z(G) = \{z \in G \mid gz = zg, \forall g \in G\}$

Lemma 1.4.5. $C_G(S)$ and $Z(G)$ are subgroups of G

Proof. Both contain e so they are nonempty

Let $a, b \in C_G(S)$ and $s \in S$

Consider:

$$\begin{aligned}
 as &= sa \\
 sab^{-1} &= asb^{-1} \\
 &= a(b^{-1}s)b \\
 &= (ab^{-1})s
 \end{aligned}$$

Thus $ab^{-1} \in C_G(S)$

Thus, by Theorem 1.4.3 $C_G(S)$ is a subgroup

$\implies Z(G) \leq G$ since the center is a special case of the centralizer

□

Example 1.4.6. Centralizers in D_4

- $C_{D_4}(e) = D_4 = C_{D_4}(r^2)$ ²
- $C_{D_4}(r) = e, r, r^2, r^3 = C_{D_4}(r^3)$
- $C_{D_4}(s) = e, s, r^2, sr^2 = C_{D_4}(sr^2)$
- $C_{D_4}(sr) = e, sr, r^2, sr^3 = C_{D_4}(sr^3)$
- $Z(D_4) = e, r^2$

²Since $r^2s = sr^2$ and $r^2r = rr^2$

1.4.2 Exercises

Example 1.4.7. Prove that if G is abelian if and only if for all $a, b \in G$ we have $(ab)^{-1} = a^{-1}b^{-1}$

Proof. ($\implies$) Assume G is abelian and let $a, b \in G$

$$\implies (ab)^{-1} = b^{-1}a^{-1} = a^{-1}b^{-1}$$

($\impliedby$) Assume for all $a, b \in G$ we have $(ab)^{-1} = a^{-1}b^{-1}$

Consider: a^{-1} and b^{-1}

$$ba = (a^{-1}b^{-1})^{-1} = ab$$

□

Example 1.4.8. Suppose G is a group of order 6 containing 7 and 19 where the operation is multiplication modulo 36. List all the elements of G .

- $7 \times 7 = 49 = 13$
- $19 \times 19 = 361 = 1$
- $7 \times 19 = 133 = 25$
- $13 \times 19 = 247 = 31$

$$G = \{1, 7, 13, 19, 25, 31\}$$

Example 1.4.9. Suppose that G is a group such that for all $a, b, c, d \in G$ such that if

$$axb = cxd$$

then $ab = cd$. Show that G is abelian

Proof. Consider: $a, b \in G$

$$aa^{-1}b = ba^{-1}a \implies ab = ba$$

□

1.5 Cyclic Groups

Definition 1.5.1. Given a group G and an element $g \in G$, the cyclic subgroup generated by g is $\langle g \rangle = \{g^m \mid m \in \mathbb{Z}\} = \{\dots, g^{-2}, g^{-1}, e, g, g^2, \dots\}$

We say G is a cyclic group if there is a $g \in G$ s.t. $G = \langle g \rangle$

$$\text{ord}(g) = |\langle g \rangle|$$

Cyclic subgroups of $\mathbb{Z}$

Example 1.5.2.

- $\langle 2 \rangle = \{2m \mid m \in \mathbb{Z}\} = \{\dots, -4, -2, 0, 2, 4, \dots\} = 2\mathbb{Z}$
- $\langle n \rangle = \{nm \mid m \in \mathbb{Z}\} = \{\dots, -2n, -n, 0, n, 2n, \dots\} = n\mathbb{Z}$

Cyclic subgroups of $\mathbb{Q}^\times$

Example 1.5.3.

- $\langle -1 \rangle = \{-1, 1\}$
- $\langle 2 \rangle = \{2^m \mid m \in \mathbb{Z}\}$

Cyclic subgroups of $\mathbb{C}^\times$

Example 1.5.4.

$$\langle i \rangle = \{i, -1, -i, 1\}$$

Cyclic subgroups of $\mathbb{Z}_{12}$

Example 1.5.5.

- $\mathbb{Z}_{12} = \langle 1 \rangle = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$
- $\langle 2 \rangle = \{0, 2, 4, 6, 8, 10\}$
- $\langle 4 \rangle = \{0, 4, 8\}$
- $\langle 3 \rangle = \{0, 3, 6, 9\}$

Cyclic subgroups of $U_9 = \{1, 2, 4, 5, 7, 8\}$

Example 1.5.6.

- $\langle 1 \rangle = \{1\}$
- $\langle 8 \rangle = \{8, 64\} = \{1, 8\}$
- $\langle 4 \rangle = \{4, 16, 64\} = \{1, 4, 7\}$
- $\langle 2 \rangle = \{2, 4, 8, 16, 32, 64\} = \{1, 2, 4, 5, 7, 8\} = U_9$

Cyclic subgroups of D_n

Example 1.5.7.

- $\langle r \rangle = \{e, r, r^2, \dots, r^{n-1}\} = Z_n^3 = C_n^4$
- $\langle s \rangle = \{e, s\}$
- $\langle sr \rangle = \{e, sr\}$

Cyclic subgroups of S_5

Example 1.5.8.

$$\langle (12)(345) \rangle^5 = \{(1), (12)(345), (354), (12), (345), (12)(354)\}$$

Theorem 1.5.9. Suppose $g \in G$ with $\text{ord}(g) = n$ and let $m \in \mathbb{N}$ then, $\langle g^m \rangle = \langle g^{\gcd(m,n)} \rangle$

Proof. Set $d = \gcd(m, n)$.

Take $x \in \langle g^m \rangle \implies x = (g^m)^k = g^{mk}$ for some $k \in \mathbb{Z}$

Also, $m = dq$ for some $q \in \mathbb{Z}$ since d is a divisor of m

$x = g^{dqk} = (g^d)^{qk} \in \langle g^d \rangle$ so $\langle g \rangle \subseteq \langle g^d \rangle$

By Lemma 0.1.2 there exists $r, s \in \mathbb{Z}$ s.t. $mr + ns = d$

Take $x \in \langle g^d \rangle \implies x = g^{dk}$ for some $k \in \mathbb{Z}$

$g^{dk} = g^{k(mr+ns)} = g^{mkr} (g^n)^{ks} = g^{mkr} \in \langle g^m \rangle$ so $\langle g^d \rangle \subseteq \langle g^m \rangle$

Thus $\langle g^m \rangle = \langle g^d \rangle$

□

³not $\mathbb{Z}$

⁴Multiplicative cyclic group of order n

⁵By Theorem 1.3.10 the $\text{lcm}(2,3)=6=\text{order}$ and thus this cyclic group will have 6 elements

Theorem 1.5.10 (Fundamental Theorem of Cyclic Groups).

- I. Every subgroup of a cyclic group is cyclic
- II. If G is a cyclic group of order n and $d \mid n$ then there is a unique subgroup $H \leq G$ with $|H| = d$

I. *Proof.* Suppose $G = \langle g \rangle$ and $H \leq G$

Case 1: $H = \{e\} = \langle e \rangle \checkmark$

Case 2: $H \neq \{e\}$. Thus, $H = \{\dots, g^{-m_2}, g^{-m_1}, e, g^{m_1}, g^{m_2}, \dots\}$ s.t. $m_i \in \mathbb{N}$

By the Well Ordering Principle, there exists a minimum m_i call this m

Since $g^m \in H \implies (g^m)^n \in H \ \forall n \in \mathbb{Z} \implies \langle g^m \rangle \subseteq H$

Take $h \in H$, write $h = g^k$ for $k \in \mathbb{Z}$

By the division algorithm (Thm. 0.1.1), there exists a $q, r \in \mathbb{Z}$ s.t.

$k = mq + r$ with $0 \leq r < m$

$g^r = (g^k)(g^{-mq})$ thus since $g^k \in H$ and $g^{-mq} \in \langle g^m \rangle \subseteq H \implies g^r \in H$

Since $0 \leq r < m$ and m is the minimum $\implies r = 0$

Thus $h = (g^m)^q \in \langle g^m \rangle \implies H \subseteq \langle g^m \rangle$

$\implies H = \langle g^m \rangle \implies H$ is cyclic □

II. *Proof.* Assume G is a cyclic group of order n and $d \mid n$

Let $G = \langle g \rangle$

Existence: Consider $\langle g^{\frac{n}{d}} \rangle$ this has d elements so $\checkmark$

Uniqueness: Let $H \leq G$, $|H| = d$ by Theorem 1.5.10.I

$\implies H = \langle g^m \rangle$ by Theorem 1.5.9 $\implies H = \langle g^{\gcd(m,n)} \rangle$

Thus $|H| = d = \frac{n}{\gcd(m,n)} \implies \gcd(m,n) = \frac{n}{d} \implies H = \langle g^{\frac{n}{d}} \rangle$ which unique describes a group □

Definition 1.5.11 (Euler's Totient Function).

For $n \in \mathbb{N}$, define $\varphi(n) = \left| \{1 \leq m \leq n \mid \gcd(m,n) = 1\} \right|$

Theorem 1.5.12.

Suppose G is a cyclic group of order n and $d \mid n$ then the number of elements of G of order d is $\varphi(d)$

Proof. Assume G is cyclic group of order n and $d \mid n$

Let $H \leq G$ be the unique subgroup s.t. $|H| = d \iff$ Theorem 1.5.10.II

Let $H = \langle h \rangle \iff$ Theorem 1.5.10.I

Let $x \in G$ have order d

Consider: $\langle x \rangle$ since x is order $d \implies |\langle x \rangle| = d$

Since H is unique $\implies \langle h \rangle = \langle x \rangle \implies x = h^m$ where $\gcd(m, d) = 1 \iff$ Theorem 1.5.9

The number of choices of m is $\varphi(d)$

□

1.5.1 Exercises

Example 1.5.13. Find all generators of $\mathbb{Z}_{15}$ and $\mathbb{Z}_{12}$

By Theorem 1.5.9 this implies all the numbers that are relatively prime to 12 and 15

Thus, the generators for $\mathbb{Z}_{15}$ are 1,2,4,7,8,11,13,14

Thus, the generators for $\mathbb{Z}_{12}$ are 1,5,7,11

Example 1.5.14. Show that $\mathbb{Q}_{\geq 0}^{\times}$ is not cyclic

Assume $\mathbb{Q}_{\geq 0}^{\times}$ is cyclic thus there exists a $q = \frac{a}{b} \in \mathbb{Q}_{\geq 0}^{\times}$ s.t. $\mathbb{Q}_{\geq 0}^{\times} = \langle q \rangle$

Consider: p a prime bigger than a, b

Let $\frac{1}{p} = \frac{a^m}{b^m}$

Case 1: $m \geq 1$

$\implies b^m = pa^m \implies p \mid b^m \implies \Leftarrow$ since we picked p to be bigger than a, b

Case 2: $m \leq -1$

$\implies a^m = pb^m \implies p \mid a^m \implies \Leftarrow$

1.6 Homomorphism and Isomorphism

Definition 1.6.1. Let $(G_1, *)$ and $(G_2, \circ)$ be groups. A map $\varphi : G_1 \rightarrow G_2$ is called a homomorphism if for all $a, b, c \in G_1$ we have $\varphi(a * b) = \varphi(a) \circ \varphi(b)$

If, in addition, φ is a bijection we say φ is an isomorphism and G_1 and G_2 are isomorphic

$$G_1 \cong G_2$$

Theorem 1.6.2. Suppose $G = \langle g \rangle$ is a cyclic group.

If $|G| = \infty$ then $G \cong \mathbb{Z}$

If $|G| = n$ then $G \cong \mathbb{Z}_n$

Proof. Assume $|G| = \infty$ Define:

$$\begin{aligned} \varphi : G &\rightarrow \mathbb{Z} \\ g^m &\mapsto m \end{aligned}$$

$$\begin{aligned} \varphi^{-1} : \mathbb{Z} &\rightarrow G \\ m &\mapsto g^m \end{aligned}$$

Let $g^a, g^b \in G$ and $c, d \in \mathbb{Z}$

Consider: $\varphi(g^a g^b) = \varphi(g^{a+b}) = a + b = \varphi(g^a) + \varphi(g^b)$

Consider: $\varphi^{-1}(c + d) = g^{c+d} = g^c g^d = \varphi^{-1}(c) + \varphi^{-1}(d)$

Consider: $\varphi \circ \varphi^{-1}(c) = \varphi(g^c) = c$

$\varphi^{-1} \circ \varphi(g^a) = \varphi^{-1}(a) = g^a$

Thus, φ is a homomorphism and bijective

$$\implies G \cong \mathbb{Z}$$

Assume $|G| = n$

$$\begin{aligned} \varphi : G &\rightarrow \mathbb{Z}_n \\ g^m &\mapsto [m] \end{aligned}$$

$$\begin{aligned} \varphi^{-1} : \mathbb{Z}_n &\rightarrow G \\ [m] &\mapsto g^m \end{aligned}$$

homomorphism: ✓

Consider: $\varphi \circ \varphi^{-1}([c]) = \varphi(g^c) = [c]$

$\varphi^{-1} \circ \varphi(g^a) = \varphi^{-1}([a]) = g^a$

Thus, φ is a homomorphism and bijective

$\implies G \cong \mathbb{Z}_n$

□

Theorem 1.6.3 (Cayley's Theorem). *Every group is isomorphic to a permutation group.*

Proof. Let G be a group and by Lemma 1.3.2 $\implies S_G$ is a group

Define:

$$\varphi : G \rightarrow S_G$$

$$g \mapsto l_g$$

$$l_g : G \rightarrow G$$

$$x \mapsto gx$$

- **Injective:** Let $x, y \in G$ s.t. $l_g(x) = l_g(y)$

$$\implies gx = gy \implies x = y$$

- **Surjective:** Let $y \in G$

$$\text{Consider: } l_g(g^{-1}y) = gg^{-1}y = y$$

Thus $l_g \in S_G$

Consider: $a, b \in G$

$$\varphi(ab)(x) = l_{ab}(x) = abx = a(bx) = l_a \circ l_b(x) = \varphi(a)\varphi(b)(x)$$

Injective: $a, b, x \in G$ s.t. $\varphi(a) = \varphi(b)$

$$l_a = l_b \implies l_a(x) = l_b(x) \quad \forall x \in G \implies ax = bx \implies ae = be \implies a = b$$

$$\implies G \cong \text{im}(G) \leq S_G$$

□

Theorem 1.6.4. Suppose $\varphi : G_1 \rightarrow G_2$ is a homomorphism then

I. $\varphi(e_1) = e_2$

II. $\forall n \in \mathbb{Z}, \varphi(g^n) = [\varphi(g)]^n$

III. $ab = ba \implies \varphi(a)\varphi(b) = \varphi(b)\varphi(a)$

IV. $\text{ord}(\varphi(g)) \mid \text{ord}(g)$

V. $H \leq G_1 \implies \varphi(H) \leq G_2$

VI. $K \leq G_2 \implies \varphi^{-1}(K) \leq G_1$

I. *Proof.* $e_2\varphi(e_1) = e_2\varphi(e_1e_1) = e_2\varphi(e_1)\varphi(e_1) \implies e_2 = \varphi(e_1)$ □

II. *Proof.* Base Case: $n = 0$ $\varphi(g^0) = \varphi(e_1) = e_2 = [\varphi(g)]^0$

Assume statement is true for $k \geq 0$ for $k \in \mathbb{N}$

Consider: $k + 1$

$$\varphi(g^{k+1}) = \varphi(g)\varphi(g^k) = \varphi(g)[\varphi(g)]^k = [\varphi(g)]^{k+1}$$

Note: $e_2 = \varphi(e_1) = \varphi(gg^{-1}) = \varphi(g)\varphi(g^{-1}) \implies [\varphi(g)]^{-1} = \varphi(g^{-1})$

This proves the statement for all negative integers since we just apply the induction on g then find its inverse □

III. *Proof.* $\varphi(a)\varphi(b) = \varphi(ab) = \varphi(ba) = \varphi(b)\varphi(a)$ □

IV. *Proof.* Set $n = \text{ord}(g)$

$$\implies \varphi(g)^n = \varphi(g^n) = \varphi(e_1) = e_2 \implies \text{ord}(\varphi(g)) \mid \text{ord}(g)$$
 □

V. *Proof.* Let $x, y \in \varphi(H)$ s.t. $\varphi(a) = x$ and $\varphi(b) = y$ for $a, b \in H$

$$\implies ab^{-1} \in H \implies \varphi(ab^{-1}) = \varphi(a)\varphi(b)^{-1} \in \varphi(H)$$

Thus, by Theorem 1.4.3 $\varphi(H) \leq G_2$ □

VI. *Proof.* Let $a, b \in \varphi^{-1}(K) \implies \varphi(a), \varphi(b) \in K$

$$\implies \varphi(ab^{-1}) = \varphi(a)\varphi(b)^{-1} \in K \implies ab^{-1} \in \varphi^{-1}(K)$$

Thus, by Theorem 1.4.3 $\varphi^{-1}(K) \leq G_1$ □

Theorem 1.6.5. *thm:isomorphismgeneral Suppose $\varphi : G_1 \rightarrow G_2$ is an isomorphism then φ^{-1} is an isomorphism*

II. G_1 is abelian $\iff G_2$ is abelian

III. G_1 is cyclic $\iff G_2$ is cyclic

IV. $\text{ord}(g) = \text{ord}(\varphi(g))$

I. $\checkmark$

II. *Proof.* Suppose G_1 is abelian and $x, y \in G_2$

$$xy = \varphi(a)\varphi(b) = \varphi(ab) = \varphi(ba) = \varphi(b)\varphi(a) = yx$$

Do same thing for other direction $\square$

III. *Proof.* Suppose G_1 is cyclic and $\langle g \rangle = G_1$

$$\text{Take } x \in G_2 \implies \varphi(a) = x$$

$$a = g^n \implies x = \varphi(g^n) = \varphi(g)^n \implies x \in \langle \varphi(g) \rangle \implies G_2 \subseteq \langle \varphi(g) \rangle$$

$$\text{Take } x \in \langle \varphi(g) \rangle \implies x = \varphi(g)^n$$

$$\implies x = \varphi(g^n) \implies x \in G_2 \implies \langle \varphi(g) \rangle \subseteq G_2$$

$$\implies G_2 = \langle \varphi(g) \rangle \implies G_2 \text{ is cyclic}$$

Do same thing for other direction $\square$

IV. *Proof.* By Theorem 1.6.4.IV $\text{ord}(g) \mid \text{ord}(\varphi(g))$ and $\text{ord}(\varphi(g)) \mid \text{ord}(g) \implies \text{ord}(g) = \text{ord}(\varphi(g))$

$\square$

1.6.1 Exercises

Example 1.6.6. Describe all homomorphisms $\varphi : \mathbb{Z}_{12} \rightarrow \mathbb{Z}_{20}$

$$\text{ord}(1) = 12 \text{ in } \mathbb{Z}_{12} \implies \text{ord}(\varphi(1)) \mid 12 \implies \text{ord}(\varphi(1)) \in \{1, 2, 3, 4, 6, 12\}$$

$$\text{ord}(m) = \frac{n}{\gcd(m,n)} \text{ in } \mathbb{Z}_n$$

$$\bullet \text{ord}(\varphi(1)) = 1 \implies \varphi(1) = 0 \implies \varphi(n) = 0$$

$$\bullet \text{ord}(\varphi(1)) = 2 \implies \varphi(1) = 10$$

$$\text{this is the only element with a gcd of 10 with 20} \implies \varphi(n) = 10n$$

$$\bullet \text{ord}(\varphi(1)) = 3, 6, 12 \text{ there is no element with these orders since these are not divisors of 10}$$

$$\bullet \text{ord}(\varphi(1)) = 4 \implies \varphi(1) = 5, 15 \implies \varphi(n) = 5n, 15n$$

Example 1.6.7. Show that $\mathbb{R} \not\cong \mathbb{R}^\times$

$$\text{Assume } \mathbb{R} \cong \mathbb{R}^\times \text{ via } \varphi : \mathbb{R}^\times \rightarrow \mathbb{R}$$

$$\text{Take } x \in \mathbb{R}^\times \text{ s.t. } \varphi(x) = -1$$

$$\text{Consider } \varphi\left(\frac{x}{2}\right)^2 = \varphi(x) = -1 \implies \varphi\left(\frac{x}{2}\right) \in \mathbb{R} \text{ with } \varphi\left(\frac{x}{2}\right)^2 = -1 \Rightarrow \Leftarrow$$

1.7 Cosets

Definition 1.7.1. Suppose G is a group $H \leq G$, and $g \in G$.

- The left coset of H with representative g is

$$gH = \{gh \mid h \in H\}$$

- The right coset of H with representative g is

$$Hg = \{hg \mid h \in H\}$$

Example 1.7.2. $G = S_3, H = \{(1), (12)\}$

$$(13)H = \{(13), (123)\} = (123)H$$

$$(23)H = \{(23), (132)\} = (132)H$$

$$H(13) = \{(13), (132)\} = H(132)$$

$$H(23) = \{(23), (123)\} = H(123)$$

Theorem 1.7.3. Suppose that G is a group, $H \leq G$, $x, y \in G$. The following are equivalent

I. $xH = yH$

II. $Hx^{-1} = Hy^{-1}$

III. $xH \subseteq yH$

IV. $y \in xH$

V. $x^{-1}y \in H$

Proof. (I. $\implies$ II.)

Assume I. Let $g \in Hx^{-1} \implies g = hx^{-1}$ for $h \in H \implies g^{-1} = xh^{-1} \in xH = yH$

$\implies g^{-1} = yj$ for $j \in H \implies g = j^{-1}y^{-1} \in Hy^{-1} \implies Hx^{-1} \subseteq Hy^{-1}$

Do the same thing for other direction thus $Hx^{-1} = Hy^{-1}$

(II. $\implies$ I.)

Assume II. Let $g \in xH \implies g = xh$ for $h \in H \implies g^{-1} = h^{-1}x^{-1} \in Hx^{-1} = Hy^{-1}$

$\implies g^{-1} = jy^{-1}$ for $j \in H \implies g = yj \in Hy \implies xH$

(III. $\implies$ IV.)

Assume III. Let $g \in xH \subseteq yH \implies g = xh = yj$ for $h, j \in H \implies y = xhj^{-1} \implies y \in xH$

(IV. $\implies$ V.)

Assume IV. $y \in xH \implies y = xh$ for $h \in H \implies x^{-1}y = h \in H$

($V \implies I.$)

Assume V. Let $g \in xH \implies g = xh$ for $h \in H \implies g = y(y^{-1}xh) \in yH$ since $y^{-1}x \in H$

Do the same thing for the other direction this $xH = yH$ □

Theorem 1.7.4. Suppose $H \leq G$. The cosets of H partition G . that is, they are disjoint and they union to G .

Proof.

Disjoint: Suppose $x, y \in G$ s.t. $xH \cap yH \neq \emptyset$

Take $g \in xH \cap yH \implies g = xh = yj$ $h, j \in H \implies x^{-1}y = hj^{-1} \in H$

Thus, by Theorem 1.7.3 $xH = yH$

Union: $\bigcup_{x \in G} xH \subseteq G \iff xH \subseteq G$

Take $g \in G$ by Theorem 1.7.3 $\implies g \in gH \implies g \in \bigcup_{x \in G} xH \implies G \subseteq \bigcup_{x \in G} xH$

$\implies G = \bigcup_{x \in G} xH$ □

Definition 1.7.5. Let G be any group with $H \leq G$.

Define the index of H in G to be the number of left(right) cosets of H in G , denote

$$[G : H]$$

Theorem 1.7.6. The number of left cosets is equal to the number of right cosets

Proof. Let $\mathfrak{L}_H = \{gH \mid g \in G\}$ and $\mathfrak{R}_H = \{Hg \mid g \in G\}$

Define:

$$f: \mathfrak{L}_H \rightarrow \mathfrak{R}_H$$

$$gH \mapsto Hg^{-1}$$

Well Defined: Suppose $xH = yH$, $f(xH) = Hx^{-1} = Hy^{-1} = f(yH)$

Injective: Suppose $f(xH) = f(yH) \implies Hx^{-1} = Hy^{-1} \implies xH = yH$

Surjective: Let $Hg \in \mathfrak{R}_H$, $f(g^{-1}H) = Hg$

Therefore, $|\mathfrak{L}_H| = |\mathfrak{R}_H|$ □